

Econ 272/Mgtecon 607 - Section 7

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1. Announcements & Discussion Questions
2. Local Average Treatment Effects (LATE)
3. Weak Instruments & Many Instruments
4. Practice Problems

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- Next Week Lectures
 - May 21 → 8-9:50am GSB C105
 - June 2, 4 no lecture (maybe section?) → May 22, 4:30-6:20pm STLC 114

LATE

- 1 What do we mean by always-takers, never-takers, compliers, and defiers?
- 2 What average treatment effect can we estimate by instrumental variables? (in words) How does the algebra work out? (formulas) Why / why not is this estimand interesting?
- 3 What is the monotonicity assumption (in words and math) and why do we need to make it

Weak Instruments

- 1 What do we mean when we say weak instruments?
- 2 What are some problems that arise when we have weak instruments?
- 3 What are some alternative confidence intervals that we can use when we have weak instruments? What are the advantages/disadvantages of using them?

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Motivation: Effect Heterogeneity

Goal: Allow for: heterogeneity in TE across units
more flexible functional forms
→ endogeneity $\neq$ correlation with residual

Challenge: Cannot ID population ATE
Can only ID ATE for sub-pop'n → compliers

- (Binary) Z : *Instrument*

- (Binary, endogenous) W : $w_i = w_i(z_i) = \begin{cases} w_i(1) & \text{if } z_i = 1 \\ w_i(0) & \text{if } z_i = 0 \end{cases}$

- Y : $y_i = y_i(w_i) = y_i(w_i(z_i)) = y_i(z_i, w_i)$

Key Assumption: Independence of Instrument

$$Z_i \perp\!\!\!\perp (Y_i(0), Y_i(1), W_i(0), W_i(1))$$

→ IV does not directly affect outcome
(nonparametric assumption)

Conditions:

■ Randomization of Z :

$$Z_i \perp (W_i(0), W_i(1), Y_i(0,0), Y_i(1,0), Y_i(0,1), Y_i(1,1))$$

■ Exclusion: → not testable, must be argued

$$\forall z', z, w: Y_i(z, w) = Y_i(z', w)$$

True Compliance Types

		$W_i(0)$ $Z_i=0$	
		0	1
$W_i(1)$ $Z_i=1$	0	never-taker	defiers
	1	complier	always-takers

Observed Compliance Types

		Z_i	
		0	1
W_i	0	never-taker/complier	never-taker/defier
	1	always-taker/defier	always-taker/complier

Monotonicity

Goal: Disambiguate type of unit from (Z_i, W_i)

Assumption: Monotonicity : $W_i(1) \geq W_i(0) \iff$ no defiers
 $\hookrightarrow = \iff$ always-/never-takers
 $\hookrightarrow > \iff$ compliers

↓
sometimes testable

more plausible
when IV incentive

		Z_i	
		0	1
W_i	0	never-taker/complier	never-taker
	1	always-taker	always-taker/complier

Conditional Means

Let proportions of compliers, never-takers, and always-takers be given by π_c, π_n, π_a .

- $E[W_i | Z_i = 0] = \pi_a$
 - $E[W_i | Z_i = 1] = \pi_a + \pi_c$
 - $E[Y_i | W_i = 0, Z_i = 0] = \frac{\pi_c}{\pi_c + \pi_n} E[Y_i(0) | c] + \frac{\pi_n}{\pi_c + \pi_n} E[Y_i(0) | n]$
 - $E[Y_i | W_i = 1, Z_i = 0] = E[Y_i(1) | a]$
 - $E[Y_i | W_i = 0, Z_i = 1] = E[Y_i(0) | n]$
 - $E[Y_i | W_i = 1, Z_i = 1] = \frac{\pi_c}{\pi_a + \pi_c} E[Y_i(1) | c] + \frac{\pi_a}{\pi_a + \pi_c} E[Y_i(1) | a]$
- ED $\pi_a, \pi_c, \pi_n = 1 - \pi_a - \pi_c$

Why only compliers? Only group with members in both treatment groups

Estimand (LATE): $LATE = E[Y_i(1) - Y_i(0) | \text{complier}]$

Estimation Strategies:

- Use previous conditional means
- ILS

- $Y \sim Z$ Slope Coefficient:** $E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]$
 $(E[Y_i(1) | c] \pi_c + E[Y_i(0) | n] \pi_n + E[Y_i(1) | a] \pi_a) = E[Y_i(1) - Y_i(0) | c] \pi_c$
 $= -(E[Y_i(0) | c] \pi_c + E[Y_i(0) | n] \pi_n + E[Y_i(1) | a] \pi_a)$
- $W \sim Z$ Slope Coefficient:** $E[W_i | Z_i = 1] - E[W_i | Z_i = 0] = \pi_c$
- Ratio:** $\tau^{zv} = \frac{E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]}{E[W_i | Z_i = 1] - E[W_i | Z_i = 0]} = E[Y_i(1) - Y_i(0) | c]$

ATE Bounds (Binary Outcomes)

Goal: Partial ID of full pop'n ATE

$$ATE = E[Y_i(1) - Y_i(0)] = E[\overset{\checkmark}{Y_i(1)} - \overset{\checkmark}{Y_i(0)} | a] \pi_a + E[\overset{\checkmark}{Y_i(1)} - \overset{\checkmark}{Y_i(0)} | n] \pi_n + E[\overset{\checkmark}{Y_i(1)} - \overset{\checkmark}{Y_i(0)} | c] \pi_c$$

→ cannot estimate $E[Y_i(0) | a]$, $E[Y_i(1) | n]$

LB: $E[Y_i(0) | a] = 1$, $E[Y_i(1) | n] = 0$

UB: $E[Y_i(0) | a] = 0$, $E[Y_i(1) | n] = 1$

$$W_i \perp (Y_i(0), Y_i(1)) \mid Z_i$$

What if we (mistakenly) included Z as a control, rather than an instrument?

DiM estimator, after adjusting for Z_i , becomes:

$$E[Y_i \mid V_i=1, Z_i=1] - E[Y_i \mid W_i=0, Z_i=1] = E[Y_i(1) \mid V_i=1, Z_i=1] - E[Y_i(0) \mid V_i=0, Z_i=1]$$

$$= \frac{\pi_u}{\pi_u + \pi_c} E[Y_i(1) \mid u] + \frac{\pi_c}{\pi_u + \pi_c} E[Y_i(1) \mid c] - E[Y_i(0) \mid u]$$

not causally interpretable

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Challenge: Standard asymptotics generally rely on nonzero correlation between
IV and endog strong 1st stage

If $\text{correl} \approx 0$, approx inaccurate even in large samples

Goal: Adjust s.e. to allow for this violation

Conventional Asymptotics (Fixed # Instruments, Fixed Parameters)

- **LIML vs. TSLS:** $\sqrt{N}(\hat{\beta}_{LIML} - \hat{\beta}_{TSLS}) \rightarrow 0$ *s.e. should be similar*

- **Just-Identified:** # IV = # endog
CI not uniformly valid, but still "decent" coverage in "reasonable" regions

(just become wider)

- **Over-Identified:** # IV > # endog
CI potentially very misleading (especially TSLS)

- **Standard Estimator:**
$$\left. \begin{aligned} x_i &= \pi_0 + \pi_1 z_i + \eta_i \\ y_i &= \alpha_0 + \alpha_1 z_i + v_i \end{aligned} \right\} \beta_1 = \frac{\alpha_1}{\pi_1} \rightarrow \hat{\beta}^{IV} = \frac{\frac{1}{N} \sum (y_i - \bar{y})(z_i - \bar{z})}{\frac{1}{N} \sum (x_i - \bar{x})(z_i - \bar{z})}$$

- **Concentration:** $\lambda = \pi_1^2 \sum \frac{(z_i - \bar{z})^2}{\sigma_\eta^2}$ *→ strength of instrument*
weak: $\lambda \rightarrow 0$

Big Idea: Let 1st stage param. depend on sample size: $\pi_{1n} = \frac{c}{\sqrt{n}}$

Then, $\lambda \rightarrow c^2 \cdot V(z_i)$

Demonstrates that for any sample size, coverage can vary a lot from desired level.

Suggested methods for uniformly valid CIs.

Goal: *Provide uniformly valid CIs*

- **Anderson-Rubin CI:**

- **Kleibergen Test:**

- **Moreira's Similar Test:**

2 problems:

- 1 Bias in standard estimators
 - 2 Misleadingly small s.e.
- } not fixable via bootstrap

Main take-away: LIML works well, but s.e. must be adjusted

2 strategies to adjust standard errors:

- 1 Bekker
- 2 REQML

→ generally recommended

Big Idea:

TSLS is not consistent in large sample, but LIML is
considers new asymptotics to derive uniformly valid CEs

$$V^{\text{Bekker}} = (V^{\text{LIML}}) \text{ (increasing \# of instruments)}$$

Random Effects Quasi Maximum Likelihood (REQML)

Big Idea: Model 1st stage coeffs. π_{ik} as iid draws from $N(\mu_\pi, \sigma_\pi^2)$

Assume joint normality, normalized Z_i , likelihood function:

$$\mathcal{L}(\beta_0, \beta_1, \pi_0, \mu_\pi, \sigma_\pi^2, \Omega)$$

6-parameter problem

TSLS: posterior mode given flat priors ($\sigma_\pi^2 = \infty$)

LDML: $\mu_\pi = 0$, Ω known

Correction REQML corrections generally larger than Bekker, because ML estimator for σ_π^2 will take into account noise in $\hat{\pi}$

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Suppose California is setting up a new job training program. To understand the impact a pilot program is set up where 10% of eligible individuals in 20 counties in California are randomly assigned to the new program. The outcome of interest is labor market status six months from randomization. Let Y_i denote the outcome, let $W_i \in \{0, 1\}$ denote the treatment, and let S_i denote the county.

Suppose 30% of the individual assigned to the program do not actually participate. Describe an instrumental variables strategy to estimate the average effect of the receipt of the training. Here you can ignore any clustering issues.

Consider the Angrist draft lottery example.

(a) Suppose that the monotonicity / no-defiers assumption does not hold. What does the two-stage-least-squares estimator estimate in that case?

(b) Suppose that the exclusion restriction does not hold for the nevertakers (but does hold for alwaystakers). Instead the effect of having a low lottery number (and being eligible for the draft) is -0.1 for the nevertakers. Suppose the fraction of nevertakers in the population is 0.7 , the fraction of compliers is 0.1 , and the fraction of alwaystakers is 0.2 (there are no defiers). What does the two-stage-least squares estimator estimate in that case in terms of average outcomes for the different compliance types.

Consider the Angrist draft lottery example.

- (a) Suppose that the monotonicity / no-defiers assumption does not hold, and there is a fraction of defiers equal to 0.01. The fraction of nevertakers is 0.7, the fraction of alwaysstakers is 0.09, and the fraction of compliers is 0.20. The average treatment effect of serving in the military for defiers is -0.1 . The tsls estimate of the effect of military service is -0.23 . What is the average effect for compliers?
- (b) Suppose you also had access to military service medical exams for all individuals. Discuss a method for using this information, and how it would improve the analysis over the simple tsls analysis.